Fiscal Year:	FY 2008	**Task Last Updated:**	FY 02/11/2008
PI Name:	McDonald, Jay M.D.		
Project Title:	Molecular Mechanisms of Osteoblastogenesis Inhibition by Modeled Microgravity		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Cell Culture (2) Translational (Countermeasure) Potential		
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Zip Code:	35294-0007	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Space Biology 01-OBPR-08
Start Date:	12/01/2003	**End Date:**	11/30/2007
No. of Post Docs:	2	**No. of PhD Degrees:**	2
No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	5
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Changed Division and Discipline/Program to FSB as of FY2006, per program changes at that time, per J. Cohen/HQ and JSC/J. Dardano and previous information fr D. Risin/JSC in Mar 2006 (jvp 6/2009) NOTE: Funding reinstated to 11/30/2007, per J. Cohen (HQ, 8/2006)		
Key Personnel Changes/Previous PI:	None.		
COI Name (Institution):			
Grant/Contract No.:	NNJ04HB27G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Bone loss and increase fractures are serious consequences of spaceflight. Understanding the mechanisms responsible for hypogravity-induced bone loss is critical for developing effective countermeasures. One of the major factors contributing to this condition is a decrease in the ability of osteoblasts bone forming cells to form healthy well-mineralized bone. Our work is focused on examining the effects of hypogravity on the ability of the human mesenchymal stem cells (the cells which gives rise to osteoblasts) to become osteoblasts. We have found that hypogravity affects these cells and forces them to become fat cells instead of bone forming cells. This is consistent with true space flights where bone loss is accompanied with increased fat cells in the bone marrow. We are presently in process of testing different therapies to countermeasure this problem which can help in preventing bone loss in astronauts as well as elderly people here on earth.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our goal is to eventually discover a new therapy to prevent and reverse bone loss.
Task Progress:	Bone loss during prolonged space flight is a serious complication which must be addressed before extended space flight (like extended stays on the moon or the trip to Mars) can be successful. Bones turn over at approximately the rate of 10% a year. An accelerated rate of bone loss over bone formation will result in an overall decrease in skeletal mass and a subsequent increase risk of fracture. The complex integration of the signals between the cells that build bones, osteoblasts, and the cells that destroy bones, osteoclasts, in the bone marrow is the focus of considerable attention by the scientific community. The successful development by NASA of the Rotary-Cell Culture System allows us to utilize our understanding of bone biology and the pathogenesis of bone disease to isolate various factors in laboratories on earth that affect bone integrity in microgravity during long-term space flight. Our research laboratory focuses on both cell types, osteoblasts and osteoclasts. We are investigating the signals whereby the cells are produced (differentiate), how these cells function (activity), and how the cells die (apoptosis). It is the integration of these processes that allows us to understand bone turnover in humans. The effects of microgravity in human studies have shown initially what we thought to be predominantly a decrease in bone formation, but subsequent studies have indicated an increase in bone breakdown as well. Therefore, the primary focus of our studies was to determine the effects of simulated microgravity on osteoblastogenesis at the intracellular signaling level, but was expanded in the last year to characterize the effect of modeled microgravity on osteoclastogenesis. Human mesenchymal stem cells which, when treated appropriately, develop into bone producing osteoblasts. If their development is altered, they may turn into fat cells, which normally occupy approximately 50% of the human bone marrow. There is also evidence from animal studies in space and from bed rest studies in humans on earth that the marrow fat cells increase under these conditions. Our studies have demonstrated that osteoblast precursors (human mesenchymal stem cells) in microgravity, instead of developing into osteoblasts developed into fat cells. The signaling process involved a decrease in the expression of collagen 1 (the main matrix protein in bone) and an increase in cell-binding integrins and subsequently in three pathways: one that involves a series of protein kinases that result in decreased osteoblastogenesis; a second path that decreases the phosphorylation of cofilin, which results in decreased stress fiber formation and altered shape of the cells; and third an enhancement of nitric oxide signaling through activation of nitric oxide synthase which will increase adipogenesis. To identify potential pathways for therapeutic intervention we fully characterized the profound anabolic effects of a commonly used drug, cyclosporine, on osteoblastogenesis. Using both pharmacological approaches and gene knockout experiments in mice, we showed that low dose cyclosporine inhibits an enzyme, calcineurin, which is responsible for enhancing osteoblastogenesis. We then focused additional efforts on developing a unique system to study osteoclastogenesis (the cells that absorb bone) in simulated microgravity. The system that we have developed using the rotary cell culture system parallels the studies on osteoblastogenesis. We have shown that modeled microgravity stimulates osteoclastogenesis, which is consistent with effects seen in humans and animals in true microgravity. Finally, the sum total of these investigations suggests that use of low dose cyclosporine or related drugs (FK506 for example) would be the ideal approach to reverse the effects of microgravity on bone loss. Through the same molecular mechanism (inhibition of calcineurin) cyclosporine inhibits osteoclastogenesis and enhances osteoblastogenesis. We anticipate that this novel approach will be successful therapy for microgravity-induced bone loss.
Bibliography Type:	Description: (Last Updated: 02/19/2008)
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